

Database Design Document

MedAssist AI

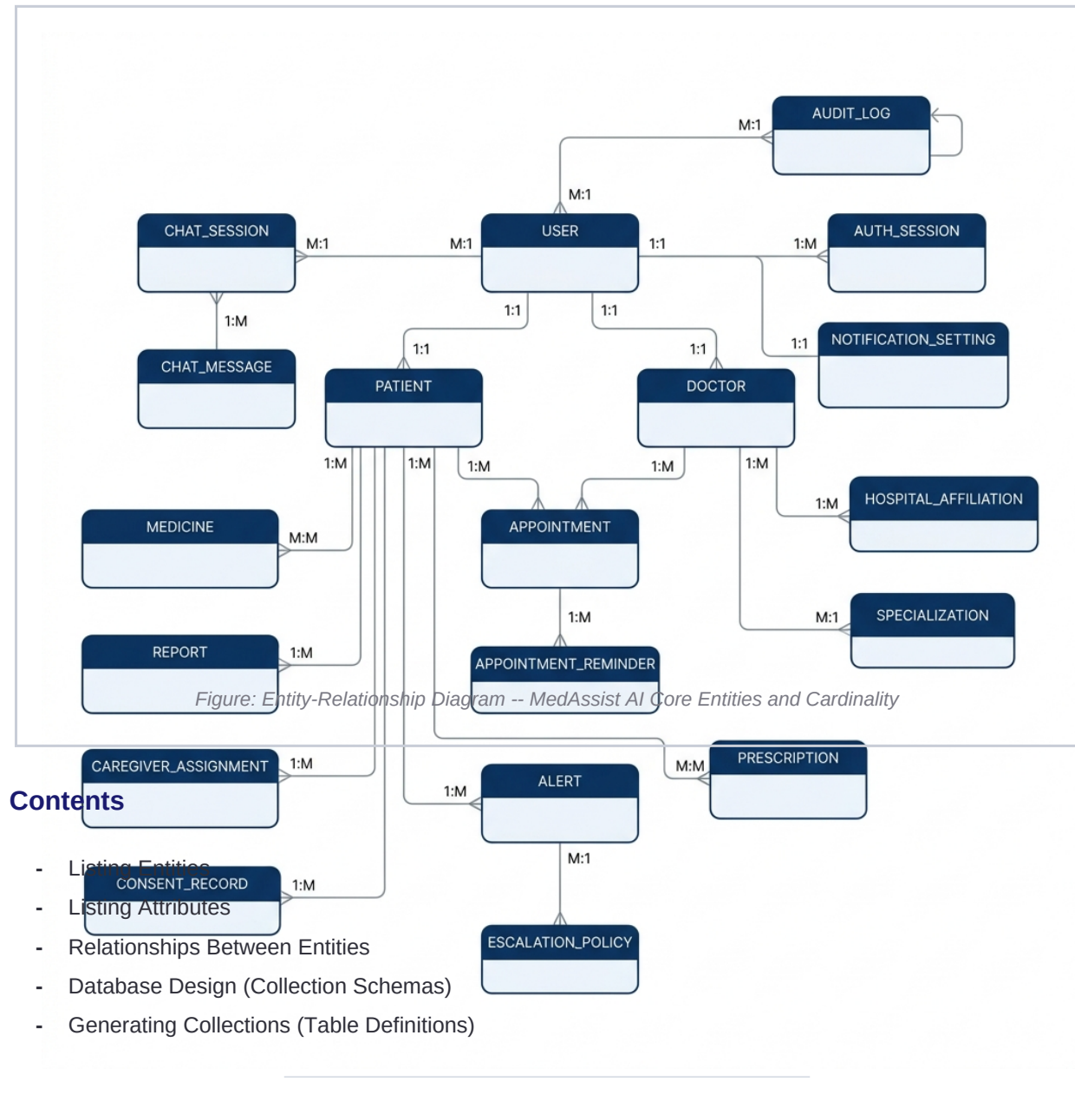
An Intelligent Healthcare Management Platform

A comprehensive database design for the MedAssist AI application, an enterprise-grade healthcare assistant system that integrates AI-powered diagnostics, medicine management, appointment scheduling, emergency alerting, and blockchain-backed audit trails across four distinct user roles: Patient, Doctor, Caregiver, and Administrator.

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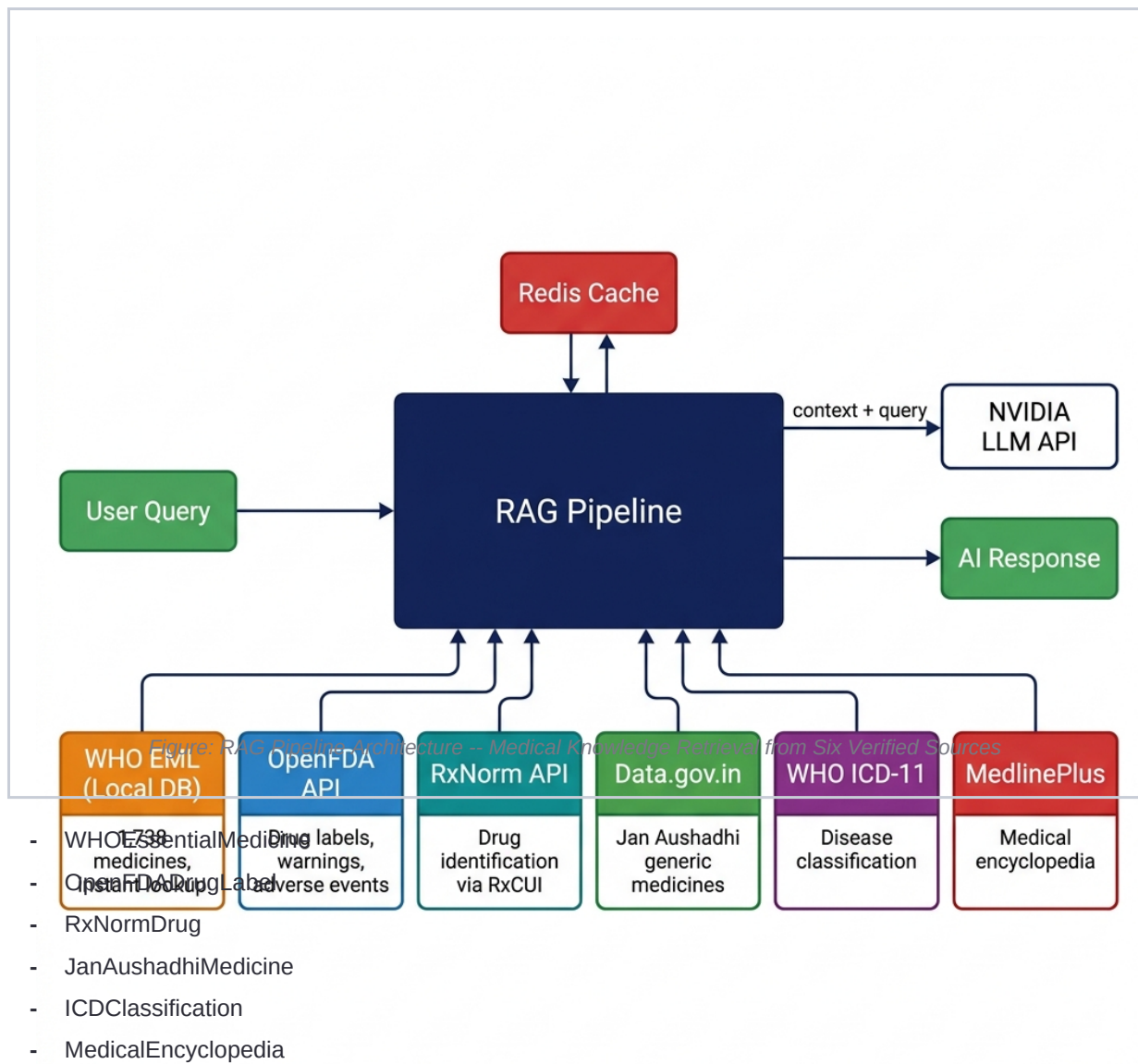
- Alert
- AlertType
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- ChatSession

- ChatMessage
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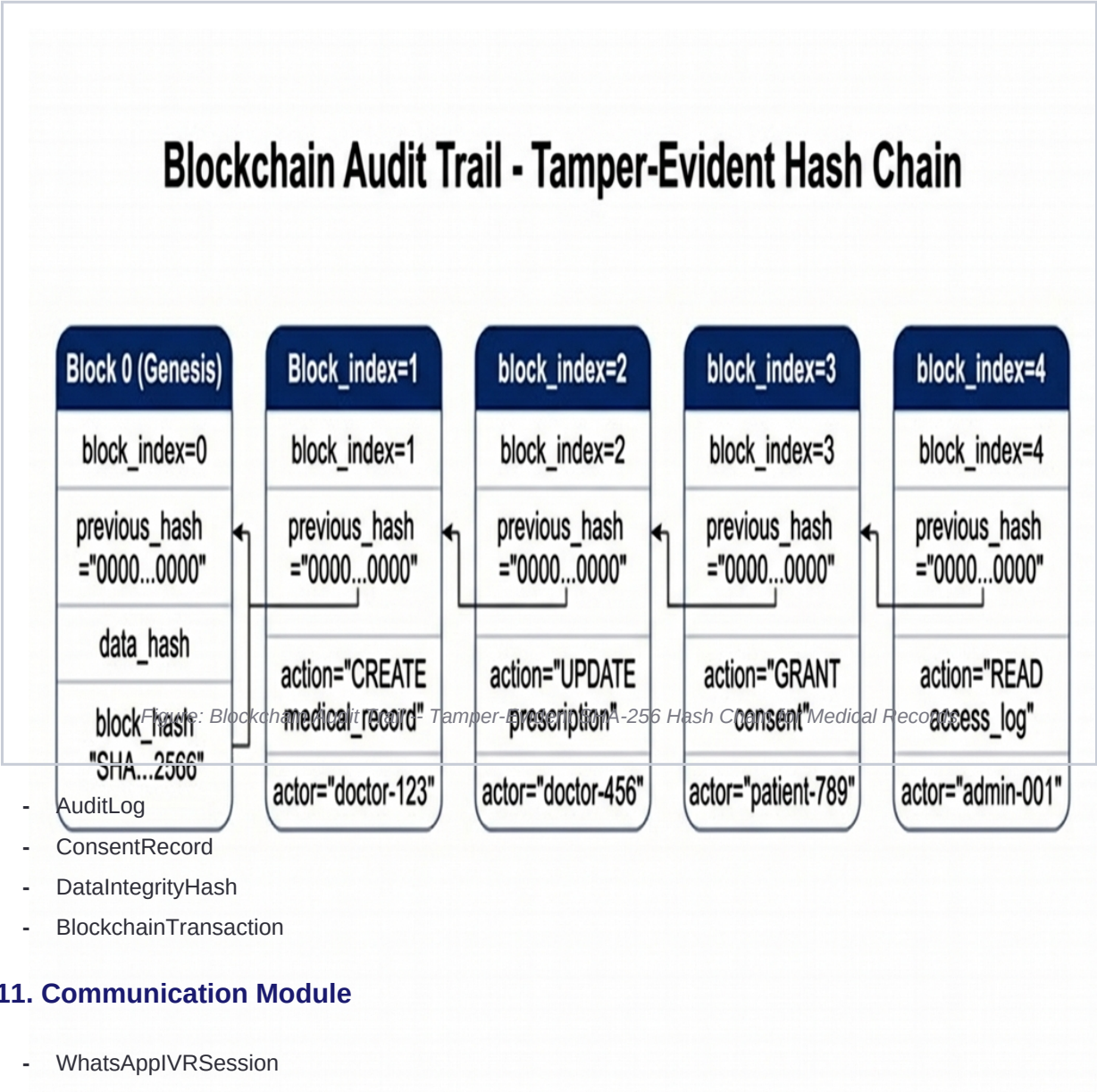
8. Medical Knowledge and RAG Module



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- Report
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- LabResult

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Listing of Attributes

1. User and Authentication Module

User

- user_id (Primary Key, auto-generated string)
- email (unique, required)
- display_name (required, minimum 1 character)

- role (enum: patient | doctor | caregiver | admin)
- phone (optional, string)
- profile_picture_url (optional, string)
- preferred_language (default: "en")
- created_at (ISO timestamp)
- updated_at (ISO timestamp)

UserRole

- role_id (Primary Key)
- role_name (patient | doctor | caregiver | admin)
- role_description (text)
- permission_level (integer, 1-4)
- is_active (boolean)

AuthSession

- session_id (Primary Key)
- user_id (Foreign Key -> User)
- firebase_token (JWT string)
- device_info (string)
- ip_address (string)
- login_timestamp (ISO timestamp)
- expiry_timestamp (ISO timestamp)
- is_valid (boolean)

SecurityCredential

- credential_id (Primary Key)
- user_id (Foreign Key -> User)
- auth_provider (enum: email | google | phone)
- hashed_password (string, for email provider)
- mfa_enabled (boolean, default false)
- last_password_change (ISO timestamp)

LanguagePreference

- preference_id (Primary Key)
- language_code (string: en | ta | hi | te | kn | ml | bn | mr | gu | pa | ur | or)
- language_name (string: English | Tamil | Hindi | Telugu | Kannada | Malayalam | Bengali | Marathi | Gujarati | Punjabi | Urdu | Odia)
- is_active (boolean)

NotificationSetting

- setting_id (Primary Key)
- user_id (Foreign Key -> User)
- push_enabled (boolean, default true)

- email_enabled (boolean, default true)
 - sms_enabled (boolean, default false)
 - whatsapp_enabled (boolean, default false)
 - quiet_hours_start (time string, optional)
 - quiet_hours_end (time string, optional)
-

2. Patient Management Module

Patient

- patient_id (Primary Key, auto-generated string)
- user_id (Foreign Key -> User)
- name (required, minimum 1 character)
- age (integer, range 0-150)
- gender (enum: Male | Female | Other)
- phone (required string)
- condition (text description of primary medical condition)
- caregiver (string, name of assigned caregiver)
- caregiver_id (Foreign Key -> User, optional)
- adherence (integer, 0-100, represents medicine adherence percentage)
- last_alert (ISO timestamp, optional)
- status (enum: Active | Stable | Needs Attention | Critical)
- blood_group (optional string)
- emergency_contact (optional string)
- created_at (ISO timestamp)
- updated_at (ISO timestamp)

PatientStatus

- status_id (Primary Key)
- status_name (enum: Active | Stable | Needs Attention | Critical)
- severity_level (integer, 1-4)
- description (text)
- color_code (hex string for UI)

CaregiverAssignment

- assignment_id (Primary Key)
 - patient_id (Foreign Key -> Patient)
 - caregiver_id (Foreign Key -> User)
 - relationship (string: parent | spouse | child | sibling | nurse | other)
 - assigned_date (ISO date)
 - is_active (boolean)
-

3. Doctor Management Module

Doctor

- doctor_id (Primary Key, auto-generated string)
- user_id (Foreign Key -> User)
- name (required, minimum 1 character)
- specialization (required string)
- hospital (required string)
- license_number (optional string)
- appointments (integer, count of scheduled appointments, default 0)
- emergencies (integer, count of active emergencies, default 0)
- rating (float, 0.0-5.0, optional)
- availability_status (enum: Available | Busy | Off-duty)
- created_at (ISO timestamp)
- updated_at (ISO timestamp)

Specialization

- specialization_id (Primary Key)
- specialization_name (string)
- department (string)
- description (text)

HospitalAffiliation

- affiliation_id (Primary Key)
- doctor_id (Foreign Key -> Doctor)
- hospital_name (string)
- hospital_address (text)
- designation (string)
- joined_date (ISO date)
- is_primary (boolean)

4. Appointment Scheduling Module

Appointment

- appointment_id (Primary Key, auto-generated string)
- patient_id (Foreign Key -> Patient)
- patient_name (denormalized string for quick lookup)
- doctor_id (Foreign Key -> Doctor)
- doctor_name (denormalized string for quick lookup)
- date (ISO date string, YYYY-MM-DD)

- time (time string, e.g., "10:00 AM")
- duration_minutes (integer, default 30)
- status (enum: Scheduled | Confirmed | Completed | Missed | Cancelled)
- notes (optional text)
- consultation_type (enum: In-Person | Telemedicine | Home Visit)
- created_at (ISO timestamp)
- updated_at (ISO timestamp)

AppointmentReminder

- reminder_id (Primary Key)
- appointment_id (Foreign Key -> Appointment)
- reminder_time (ISO timestamp)
- notification_channel (enum: push | sms | email | whatsapp)
- is_sent (boolean)
- sent_at (ISO timestamp, optional)

5. Medicine and Prescription Module

Medicine

- medicine_id (Primary Key, auto-generated string)
- name (required, minimum 1 character)
- dosage (required string, e.g., "500mg")
- frequency (required string, e.g., "Twice daily")
- patient_id (Foreign Key -> Patient)
- reminder_times (array of time strings, e.g., ["08:00", "20:00"])
- notes (optional text)
- prescribed_by (Foreign Key -> Doctor, optional)
- start_date (ISO date, optional)
- end_date (ISO date, optional)
- is_active (boolean, default true)
- created_at (ISO timestamp)
- updated_at (ISO timestamp)

Prescription

- prescription_id (Primary Key)
- patient_id (Foreign Key -> Patient)
- doctor_id (Foreign Key -> Doctor)
- medicines (array of embedded Medicine references)
- diagnosis (text)
- prescription_date (ISO date)
- valid_until (ISO date)

- digital_signature_hash (string)
-

6. Alert and Emergency Module

Alert

- alert_id (Primary Key, auto-generated string)
- type (enum: Emergency | Missed Medicine | Warning)
- patient_name (string)
- patient_id (Foreign Key -> Patient, optional)
- symptom (text description)
- status (enum: unread | read | Resolved)
- severity (enum: Low | Medium | High | Critical)
- time (ISO timestamp, optional)
- assigned_doctor_id (Foreign Key -> Doctor, optional)
- resolved_by (Foreign Key -> User, optional)
- resolution_notes (text, optional)
- created_at (ISO timestamp)
- updated_at (ISO timestamp)

EscalationPolicy

- policy_id (Primary Key)
 - alert_type (string)
 - severity_threshold (enum: Medium | High | Critical)
 - escalation_target (enum: doctor | admin | emergency_services)
 - response_timeout_minutes (integer)
 - auto_escalate (boolean)
-

7. AI Health Assistant Module

ChatSession

- session_id (Primary Key)
- user_id (Foreign Key -> User)
- language (string, language code)
- role (string: patient | doctor | caregiver)
- started_at (ISO timestamp)
- ended_at (ISO timestamp, optional)
- message_count (integer)
- emergency_detected (boolean, default false)

ChatMessage

- message_id (Primary Key)
- session_id (Foreign Key -> ChatSession)
- role (enum: user | assistant)
- text (text content)
- timestamp (ISO timestamp)
- rag_sources_used (array of strings, optional)
- emergency_flag (boolean, default false)

VoiceSession

- voice_session_id (Primary Key)
- user_id (Foreign Key -> User)
- language (string, language code)
- speech_to_text_input (text)
- ai_response_text (text)
- text_to_speech_output_url (string, optional)
- duration_seconds (float)
- created_at (ISO timestamp)

AIQueryLog

- query_log_id (Primary Key)
- user_id (Foreign Key -> User)
- query_text (text)
- response_text (text)
- model_used (string: meta/llama-3.1-8b-instruct | openai/gpt-oss-120b)
- rag_context_length (integer)
- sources_count (integer)
- response_time_ms (integer)
- emergency_detected (boolean)
- timestamp (ISO timestamp)

8. Medical Knowledge and RAG Module

WHOEssentialMedicine

- eml_id (Primary Key)
- medicine_name (string)
- generic_name (string)
- category (string)
- formulation (string)
- strength (string)
- indications (text)
- contraindications (text)

- who_model_list_edition (integer)

OpenFDADrugLabel

- fda_label_id (Primary Key)
- brand_name (string)
- generic_name (string)
- indications_and_usage (text)
- warnings (text)
- dosage_and_administration (text)
- adverse_reactions (text)
- contraindications (text)
- api_key_authenticated (boolean)
- last_fetched_at (ISO timestamp)

RxNormDrug

- rxcui (Primary Key, string)
- drug_name (string)
- drug_type (string)
- properties (embedded object)
- last_fetched_at (ISO timestamp)

JanAushadhiMedicine

- jan_aushadhi_id (Primary Key)
- generic_name (string)
- medicine_name (string)
- mrp (decimal, in INR)
- unit_size (string)
- manufacturer (string)
- data_gov_resource_id (string)
- last_fetched_at (ISO timestamp)

ICDClassification

- icd_code (Primary Key, string)
- title (string)
- definition (text)
- parent_code (string, optional)
- chapter (string)
- source (string: WHO ICD-11 API)
- last_fetched_at (ISO timestamp)

9. Health Reports Module

Report

- report_id (Primary Key, auto-generated string)
- patient_id (Foreign Key -> Patient)
- type (enum: Health Summary | Lab Results | Adherence Report)
- data (flexible JSON payload containing health metrics)
- generated_at (ISO timestamp)
- generated_by (Foreign Key -> User, optional)
- created_at (ISO timestamp)
- updated_at (ISO timestamp)

HealthMetric

- metric_id (Primary Key)
- patient_id (Foreign Key -> Patient)
- metric_name (string: blood_sugar | blood_pressure | heart_rate | temperature | spo2 | weight)
- value (string)
- unit (string: mg/dL | mmHg | bpm | degF | % | kg)
- recorded_at (ISO timestamp)
- recorded_by (enum: patient | device | doctor)

10. Blockchain Audit Trail Module

AuditLog

- audit_id (Primary Key, auto-generated string)
- record_type (enum: medical_record | consent | prescription | access_log)
- record_id (string, reference to source record)
- actor_id (Foreign Key -> User)
- action (enum: create | read | update | delete | verify)
- data_hash (SHA-256 string)
- previous_hash (SHA-256 string, linking to previous block)
- block_hash (SHA-256 string, hash of current block)
- timestamp (ISO timestamp)
- block_index (integer, sequential block number)
- verified (boolean, default true)
- metadata (flexible JSON object, optional)

ConsentRecord

- consent_id (Primary Key, auto-generated string)
- patient_id (Foreign Key -> Patient)
- consent_type (enum: data_sharing | treatment | research)
- granted_to (string, doctor ID or organization identifier)

- details (text, optional description)
- consent_hash (SHA-256 string)
- transaction_id (blockchain transaction identifier)
- timestamp (ISO timestamp)
- is_active (boolean, default true)
- revoked_at (ISO timestamp, optional)
- verified (boolean, default true)

DataIntegrityHash

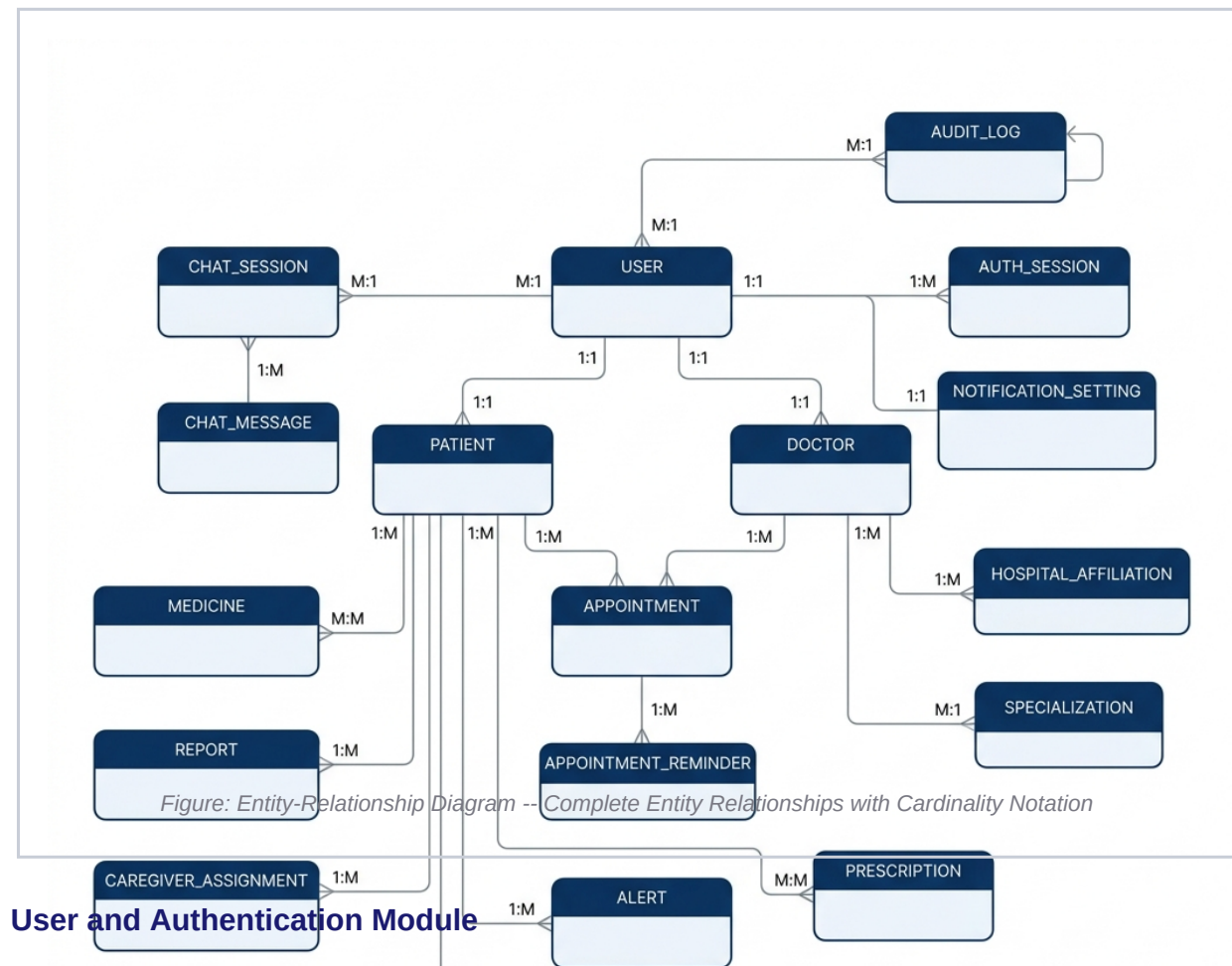
- hash_id (Primary Key)
- record_type (string)
- record_id (string)
- current_hash (SHA-256 string)
- computed_at (ISO timestamp)
- is_valid (boolean)
- last_verified_at (ISO timestamp)

11. Communication Module

WhatsAppIVRSession

- ivr_session_id (Primary Key)
- user_id (Foreign Key -> User)
- phone_number (string)
- language (string, language code)
- session_type (enum: medicine_reminder | appointment_booking | emergency_report | health_query)
- interaction_log (array of embedded message objects)
- started_at (ISO timestamp)
- ended_at (ISO timestamp, optional)
- status (enum: active | completed | abandoned)

Relationships Between Entities



User and Authentication Module

- User -----> User
 - Cardinality: 1 to Many (one role supports multiple users)
- User -----> AuthSession
 - Cardinality: 1 to Many (one user holds multiple sessions)
- User -----> SecurityCredential
 - Cardinality: 1 to Many (one user may have multiple auth providers)
- User -----> NotificationSetting
 - Cardinality: 1 to 1 (one user has one notification configuration)

Patient Management Module

- User -----> Patient
 - Cardinality: 1 to 1 (one user account maps to one patient profile)
- PatientStatus -----> Patient
 - Cardinality: 1 to Many (one status category covers multiple patients)
- Patient -----> CaregiverAssignment
 - Cardinality: 1 to Many (one patient may have multiple caregivers)
- User (Caregiver) -----> CaregiverAssignment
 - Cardinality: 1 to Many (one caregiver may be assigned to multiple patients)

Doctor Management Module

- User -----> Doctor
 - Cardinality: 1 to 1 (one user account maps to one doctor profile)
- Specialization -----> Doctor
 - Cardinality: 1 to Many (one specialization covers multiple doctors)
- Doctor -----> HospitalAffiliation
 - Cardinality: 1 to Many (one doctor affiliates with multiple hospitals)

Appointment Scheduling Module

- Patient -----> Appointment
 - Cardinality: 1 to Many (one patient books multiple appointments)
- Doctor -----> Appointment
 - Cardinality: 1 to Many (one doctor serves multiple appointments)
- Appointment -----> AppointmentReminder
 - Cardinality: 1 to Many (one appointment triggers multiple reminders)

Medicine and Prescription Module

- Patient -----> Medicine
 - Cardinality: 1 to Many (one patient takes multiple medicines)
- Doctor -----> Prescription
 - Cardinality: 1 to Many (one doctor writes multiple prescriptions)
- Patient -----> Prescription
 - Cardinality: 1 to Many (one patient receives multiple prescriptions)
- Prescription -----> Medicine
 - Cardinality: Many to Many (bridge: Prescription_Medicine)

Alert and Emergency Module

- Patient -----> Alert
 - Cardinality: 1 to Many (one patient may generate multiple alerts)
- Doctor -----> Alert
 - Cardinality: 1 to Many (one doctor may be assigned multiple alerts)
- Alert -----> EscalationPolicy
 - Cardinality: Many to 1 (multiple alerts follow one escalation policy)

AI Health Assistant Module

- User -----> ChatSession
 - Cardinality: 1 to Many (one user initiates multiple chat sessions)
- ChatSession -----> ChatMessage

- Cardinality: 1 to Many (one session contains multiple messages)
- User -----> VoiceSession
 - Cardinality: 1 to Many (one user initiates multiple voice sessions)
- User -----> AIQueryLog
 - Cardinality: 1 to Many (one user generates multiple query logs)

Medical Knowledge and RAG Module

- WHOEssentialMedicine -----> ChatMessage (via RAG context)
 - Cardinality: Many to Many (bridge: RAGContextSource)
- OpenFDADrugLabel -----> ChatMessage (via RAG context)
 - Cardinality: Many to Many (bridge: RAGContextSource)
- RxNormDrug -----> ChatMessage (via RAG context)
 - Cardinality: Many to Many (bridge: RAGContextSource)

Health Reports Module

- Patient -----> Report
 - Cardinality: 1 to Many (one patient has multiple reports)
- Patient -----> HealthMetric
 - Cardinality: 1 to Many (one patient records multiple metrics)

Blockchain Audit Trail Module

- User -----> AuditLog
 - Cardinality: 1 to Many (one user generates multiple audit entries)
- Patient -----> ConsentRecord
 - Cardinality: 1 to Many (one patient grants multiple consents)
- AuditLog -----> AuditLog (self-referencing chain)
 - Cardinality: 1 to 1 (each block links to its previous block via previous_hash)

Communication Module

- User -----> WhatsAppIVRSession
 - Cardinality: 1 to Many (one user conducts multiple IVR sessions)
- Appointment -----> WhatsAppIVRSession
 - Cardinality: 1 to 1 (one appointment may originate from one IVR session)

Cross-Module Relationships

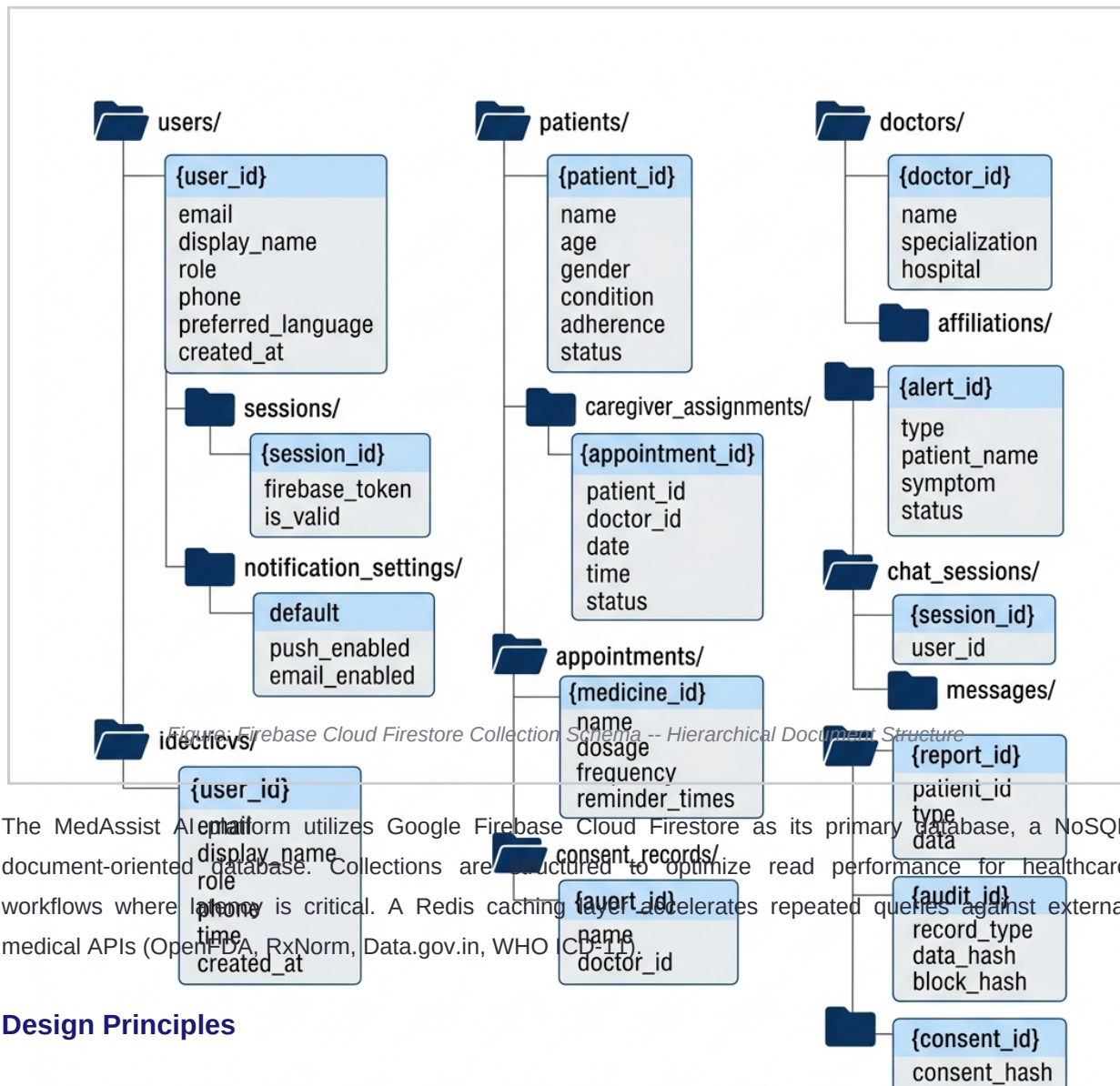
- Patient -----> Doctor
 - Cardinality: Many to Many (bridge: Appointment)
- Medicine -----> Alert
 - Cardinality: 1 to Many (one missed medicine triggers one alert)

- ChatSession -----> Alert
 - Cardinality: 1 to Many (an AI session detecting emergency generates an alert)
- AuditLog -----> All Entities
 - Cardinality: Many to 1 (any entity modification is logged as an audit block)

Notation Guide:

- 1 to Many: One record on the left relates to multiple records on the right
- Many to Many: Multiple records on both sides (requires a bridge/junction collection)
- ---> denotes the direction of the foreign key reference

Database Design



The MedAssist AI platform utilizes Google Firebase Cloud Firestore as its primary database, a NoSQL document-oriented database. Collections are structured to optimize read performance for healthcare workflows where latency is critical. A Redis caching layer accelerates repeated queries against external medical APIs (OpenFDA, RxNorm, Data.gov.in, WHO ICD-11).

Design Principles

1. Denormalization for Speed: Patient and doctor names are stored alongside appointment records to eliminate join operations during dashboard rendering.
2. Blockchain Immutability: Every data modification event produces a tamper-evident audit block with SHA-256 hashing and chain linking.
3. Flexible Schema: The Report entity uses a schemaless `data` field to accommodate diverse health metrics without schema migration.
4. Cache-First Architecture: External API responses from OpenFDA, RxNorm, and Data.gov.in are cached in Redis with configurable TTL (default 24 hours) to minimize external dependency.
5. Role-Based Segregation: Firestore security rules enforce that patients can only read their own records, doctors access assigned patient records, and administrators have full visibility.

1. User and Authentication Collections

```
Collection: users
|---- Document: {user_id}
|   |---- email: string (unique)
|   |---- display_name: string
|   |---- role: string (patient | doctor | caregiver | admin)
|   |---- phone: string (optional)
|   |---- preferred_language: string (default "en")
|   |---- created_at: timestamp
|   |---- updated_at: timestamp
|
|---- Subcollection: sessions
|   |---- Document: {session_id}
|       |---- firebase_token: string
|       |---- device_info: string
|       |---- login_timestamp: timestamp
|       |---- is_valid: boolean
|
|---- Subcollection: notification_settings
|   |---- Document: default
|       |---- push_enabled: boolean
|       |---- email_enabled: boolean
|       |---- sms_enabled: boolean
|       |---- whatsapp_enabled: boolean
```

2. Patient Management Collections

```
Collection: patients
|---- Document: {patient_id}
|   |---- user_id: string (ref -> users)
|   |---- name: string
|   |---- age: number (0-150)
|   |---- gender: string (Male | Female | Other)
|   |---- phone: string
```

```
| |---- condition: string
| |---- caregiver: string (optional)
| |---- caregiver_id: string (ref -> users, optional)
| |---- adherence: number (0-100)
| |---- last_alert: timestamp (optional)
| |---- status: string (Active | Stable | Needs Attention | Critical)
| |---- created_at: timestamp
| |---- updated_at: timestamp
|
|---- Subcollection: caregiver_assignments
  |---- Document: {assignment_id}
    |---- caregiver_id: string (ref -> users)
    |---- relationship: string
    |---- assigned_date: date
    |---- is_active: boolean
```

3. Doctor Management Collections

```
Collection: doctors
|---- Document: {doctor_id}
| |---- user_id: string (ref -> users)
| |---- name: string
| |---- specialization: string
| |---- hospital: string
| |---- appointments: number (default 0)
| |---- emergencies: number (default 0)
| |---- created_at: timestamp
| |---- updated_at: timestamp
|
|---- Subcollection: affiliations
  |---- Document: {affiliation_id}
    |---- hospital_name: string
    |---- designation: string
    |---- is_primary: boolean
```

4. Appointment Scheduling Collections

```
Collection: appointments
|---- Document: {appointment_id}
| |---- patient_id: string (ref -> patients)
| |---- patient_name: string (denormalized)
| |---- doctor_id: string (ref -> doctors)
| |---- doctor_name: string (denormalized)
| |---- date: string (YYYY-MM-DD)
| |---- time: string (e.g., "10:00 AM")
| |---- status: string (Scheduled | Confirmed | Completed | Missed | Cancelled)
| |---- created_at: timestamp
| |---- updated_at: timestamp
```

5. Medicine and Prescription Collections

```
Collection: medicines
`---- Document: {medicine_id}
  |---- name: string
  |---- dosage: string (e.g., "500mg")
  |---- frequency: string (e.g., "Twice daily")
  |---- patient_id: string (ref -> patients)
  |---- reminder_times: array of strings (e.g., ["08:00", "20:00"])
  |---- notes: string (optional)
  |---- is_active: boolean (default true)
  |---- created_at: timestamp
  `---- updated_at: timestamp
```

6. Alert and Emergency Collections

```
Collection: alerts
`---- Document: {alert_id}
  |---- type: string (Emergency | Missed Medicine | Warning)
  |---- patient_name: string
  |---- symptom: string
  |---- status: string (unread | read | Resolved)
  |---- time: timestamp (optional)
  |---- created_at: timestamp
  `---- updated_at: timestamp
```

7. AI Health Assistant Collections

```
Collection: chat_sessions
|---- Document: {session_id}
|  |---- user_id: string (ref -> users)
|  |---- language: string
|  |---- role: string
|  |---- started_at: timestamp
|  |---- ended_at: timestamp (optional)
|  `---- emergency_detected: boolean
|
`---- Subcollection: messages
  `---- Document: {message_id}
    |---- role: string (user | assistant)
    |---- text: string
    |---- timestamp: timestamp
    `---- rag_sources_used: array of strings
```

8. Health Reports Collections

```
Collection: reports
`---- Document: {report_id}
  |---- patient_id: string (ref -> patients)
  |---- type: string (Health Summary | Lab Results | Adherence Report)
  |---- data: map (flexible JSON: { blood_sugar: "120 mg/dL", bp: "130/85", ... })
  |---- generated_at: timestamp
  |---- created_at: timestamp
  `---- updated_at: timestamp
```

9. Blockchain Audit Trail Collections

```
Collection: audit_logs
`---- Document: {audit_id}
  |---- record_type: string (medical_record | consent | prescription | access_log)
  |---- record_id: string
  |---- actor_id: string (ref -> users)
  |---- action: string (create | read | update | delete | verify)
  |---- data_hash: string (SHA-256)
  |---- previous_hash: string (SHA-256)
  |---- block_hash: string (SHA-256)
  |---- timestamp: timestamp
  |---- block_index: number (sequential)
  |---- verified: boolean (default true)
  `---- metadata: map (optional)

Collection: consent_records
`---- Document: {consent_id}
  |---- patient_id: string (ref -> patients)
  |---- consent_type: string (data_sharing | treatment | research)
  |---- granted_to: string
  |---- details: string (optional)
  |---- consent_hash: string (SHA-256)
  |---- transaction_id: string
  |---- timestamp: timestamp
  `---- verified: boolean (default true)
```

10. Cache Layer (Redis)

```
Namespace: rag:{source}:{query}
|---- rag:openfda:{drug_name}      -> cached drug label text (TTL: 24h)
|---- rag:openfda_adverse:{drug_name} -> cached adverse events (TTL: 24h)
|---- rag:rxnorm:{drug_name}      -> cached RxCUI + properties (TTL: 24h)
|---- rag:datagov_med:{query}      -> cached Jan Aushadhi results (TTL: 24h)
|---- rag:datagov_hosp:{location}  -> cached hospital directory (TTL: 24h)
|---- rag:medline:{query}         -> cached encyclopedia data (TTL: 24h)
```

```
`---- rate_limit:{client_id}          -> sorted set for sliding window rate limiter
```

Generating Collections (Table Definitions)

The following Firestore security rules and index definitions ensure proper access control and query optimization.

Firestore Security Rules (Summary)

```
rules_version = '2';
service cloud.firestore {
  match /databases/{database}/documents {

    // Users: authenticated users can read own profile
    match /users/{userId} {
      allow read: if request.auth.uid == userId;
      allow write: if request.auth.uid == userId || isAdmin();
    }

    // Patients: patients read own, doctors read assigned
    match /patients/{patientId} {
      allow read: if isOwnerOrAssigned(patientId);
      allow write: if isDoctor() || isAdmin();
    }

    // Appointments: both parties can read
    match /appointments/{appointmentId} {
      allow read: if isAppointmentParticipant(resource);
      allow write: if isDoctor() || isAdmin();
    }

    // Medicines: patient reads own, doctor writes
    match /medicines/{medicineId} {
      allow read: if resource.data.patient_id == request.auth.uid;
      allow write: if isDoctor() || isAdmin();
    }

    // Alerts: doctors and admins full access
    match /alerts/{alertId} {
      allow read, write: if isDoctor() || isAdmin() || isCaregiver();
    }

    // Audit Logs: read-only for admins, write by system
    match /audit_logs/{logId} {
      allow read: if isAdmin();
      allow write: if false; // system-only writes via Admin SDK
    }
  }
}
```



```
// Consent Records: patient reads own
match /consent_records/{consentId} {
  allow read: if resource.data.patient_id == request.auth.uid || isAdmin();
  allow write: if false; // system-only writes
}
}
```

Composite Index Definitions

Collection	Fields Indexed	Query Supported
patients	status (ASC), adherence (ASC)	Filter patients by status sorted by adherence
patients	caregiver_id (ASC), status (ASC)	Caregiver viewing their assigned patients by status
appointments	doctor_id (ASC), date (ASC)	Doctor viewing appointments for a specific date
appointments	patient_id (ASC), status (ASC)	Patient viewing their appointments filtered by status
appointments	date (ASC), status (ASC)	Admin viewing all appointments on a date by status
medicines	patient_id (ASC), is_active (ASC)	Listing active medicines for a patient
alerts	status (ASC), created_at (DESC)	Dashboard showing unread alerts (newest first)
alerts	type (ASC), status (ASC)	Filtering alerts by type and resolution status
audit_logs	actor_id (ASC), timestamp (DESC)	Viewing audit trail for a specific user
audit_logs	record_type (ASC), timestamp (DESC)	Filtering audit logs by record type
consent_records	patient_id (ASC), consent_type (ASC)	Patient viewing their consent grants by type
reports	patient_id (ASC), type (ASC)	Patient viewing reports filtered by type
chat_sessions	user_id (ASC), started_at (DESC)	Listing recent chat sessions for a user

Data Volume Estimates

Collection	Estimated Records (Year 1)	Average Document Size	Storage Estimate
users	10,000	0.5 KB	5 MB
patients	8,000	1 KB	8 MB
doctors	500	0.8 KB	0.4 MB
appointments	50,000	0.6 KB	30 MB
medicines	25,000	0.5 KB	12.5 MB
alerts	30,000	0.4 KB	12 MB
audit_logs	200,000	0.8 KB	160 MB
consent_records	15,000	0.5 KB	7.5 MB
reports	20,000	2 KB	40 MB

Collection	Estimated Records (Year 1)	Average Document Size	Storage Estimate
chat_sessions	100,000	0.3 KB	30 MB
chat_messages	500,000	0.4 KB	200 MB
Total	**~958,000**	**~505 MB**	

Summary

This database design document defines the complete data architecture for the MedAssist AI healthcare platform. The design spans 11 functional modules comprising 40+ entity types stored across Firebase Cloud Firestore collections with Redis caching for external medical API responses. Key architectural decisions include denormalized fields for dashboard performance, blockchain-style hash chaining for tamper-proof audit trails, flexible JSON payloads for health reports, and role-based access control enforced at the database security rule level. The schema supports multi-language interactions in 12 Indian languages, real-time emergency alerting, AI-powered medical consultations backed by six verified knowledge sources (WHO EML, OpenFDA, RxNorm, Data.gov.in Jan Aushadhi, WHO ICD-11, and MedlinePlus), and complete HIPAA-aligned consent management through cryptographic verification.